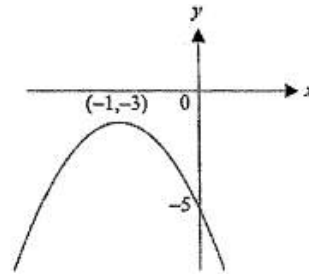


1. The function $f(x) = x^2$ is transformed into a new function whose graph is shown in the diagram below. Find the equation of the new function in the form $g(x) = k f(x+b) + c$ for some constants k , b and c .



3

The curve is reflected in the x -axis, dilated vertically then translated 1 unit to the left and down by 3 units.

$$\therefore g(x) = k(x+1) - 3$$

$$g(0) = -5$$

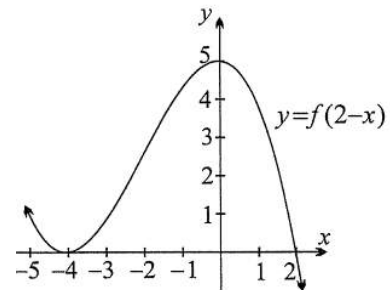
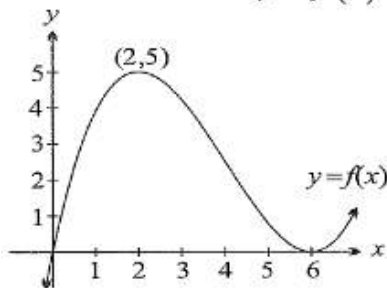
$$\therefore k(0+1) - 3 = -5$$

$$k - 3 = -5$$

$$k = -2$$

$$\therefore g(x) = -2(x+1)^2 - 3$$

2. The sketch shows $y = f(x)$. Draw a sketch of $f(2-x)$.



2

3. The graph of the function $f(x) = 3x^{\frac{5}{2}}$ is reflected in the x axis and then translated 3 units to the right and 4 units down. The equation of the transformed function is

1

A. $y = 3(-x-3)^{\frac{5}{2}} - 4$

C. $y = -3(x+3)^{\frac{5}{2}} - 4$

B. $y = -3(x-3)^{\frac{5}{2}} - 4$

D. $y = 3(-x+3)^{\frac{5}{2}} - 4$

$$f(x) = 3x^{\frac{5}{2}}$$

after reflection: $y = -3x^{\frac{5}{2}}$

after translation 3 units right: $y = -3(x-3)^{\frac{5}{2}}$

after translation 4 units down: $y = -3(x-3)^{\frac{5}{2}} - 4$

4. Describe the transformations applied to $y = x^2$ in order to give the transformed function of $y = (4x+2)^2$.

2

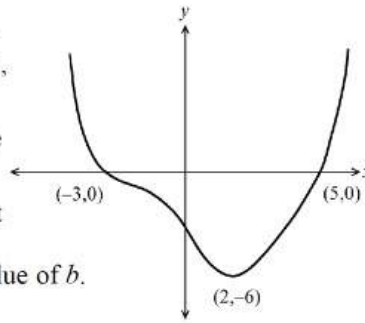
$$y = x^2 \quad \text{OR} \quad y = (2(2x+1))^2$$

$$y = (4x+2)^2 \quad y = 4(2x+1)^2$$

$$y = \left(\frac{x+\frac{1}{2}}{\frac{1}{4}}\right)^2 \quad \frac{y}{4} = \left(\frac{x+\frac{1}{2}}{\frac{1}{2}}\right)^2$$

dilate horizontally by a factor of $\frac{1}{4}$, translate to the left by $\frac{1}{2}$
OR $y = \left(\frac{x+2}{4}\right)^2$ translate to left by 2, then dilate by $\frac{1}{4}$.

5. The diagram shows a sketch of the graph of $y = f(x)$. The graph passes through the points $(-3, 0)$ and $(5, 0)$, and has a minimum turning point at $(2, -6)$.



- (a) The graph of $y = 4f(x+a)$ passes through the origin. Write down the possible values of a . 2
- (b) The curve $y = f(x)$ is now transformed so that the y -coordinate of the stationary point on the graph of $y = bf(x+2)$ is 4. Write down the value of b . 1

(a) $a = -3, 5$

(b) $bx - 6 = 4$
 $b = -\frac{2}{3}$

6. The graph of a function f is obtained from the graph of the function g with rule $g(x) = \sqrt{2x-5}$ by a reflection in the x -axis, followed by a horizontal dilation from the y -axis by a factor of $\frac{1}{2}$. Which one of the following is the rule for the function f ? 1

A. $f(x) = \sqrt{5-4x}$

B. $f(x) = -\sqrt{x-5}$

C. $f(x) = \sqrt{x+5}$

D. $f(x) = -\sqrt{4x-5}$

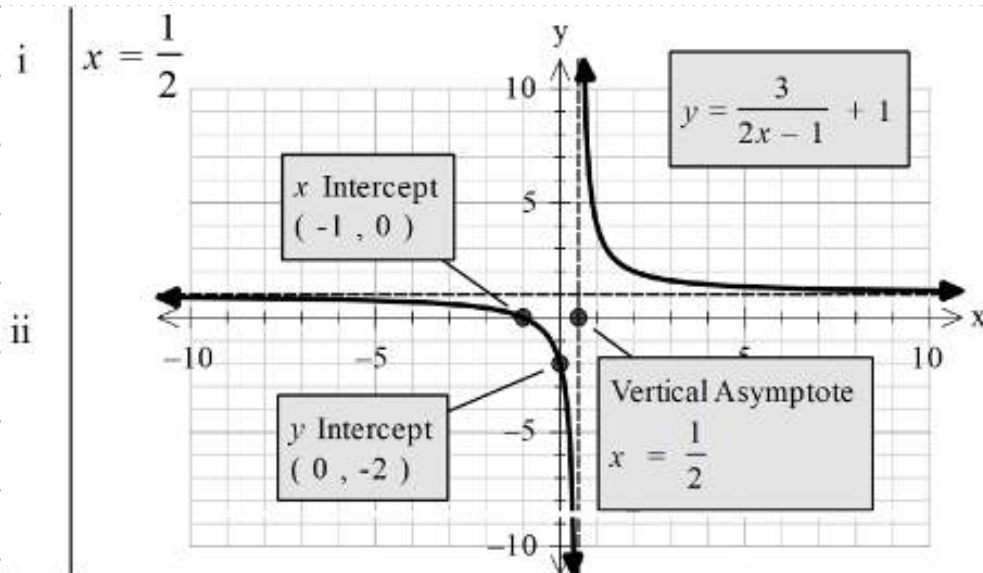
• $y = -f(x) = -\sqrt{2x-5}$

• $y = -f(2x) = -\sqrt{2(2x)-5}$
 $y = -\sqrt{4x-5}$

7. Given the function $g(x) = \frac{3}{2x-1} + 1$

- i. What is the equation of the vertical asymptote? 1
- ii. Sketch the function $g(x)$ in the space below, labelling asymptotes and intercepts. 2
- iii. The function $g(x)$ was derived from applying transformations to the function $f(x) = \frac{1}{x}$. 1

What was the horizontal translation applied to achieve this transformation?

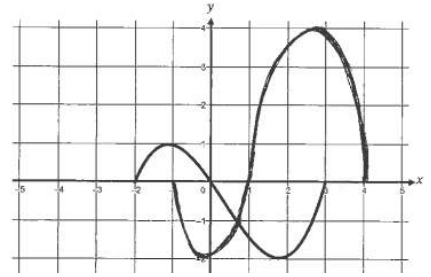
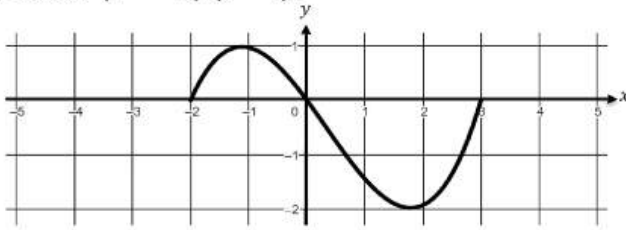


- iii. Right $\frac{1}{2}$ unit (if follows dilation); Right 1 unit if precedes dilation,

8. The graph below shows $y = f(x)$.

2

On the same graph sketch $y = -2f(x-1)$.



9. The parabola $y = x^2 - 8x + 17$ is dilated vertically by a factor of 2 and reflected about the y-axis.

3

Find the equation of the transformed parabola in the form $y = a(x+b)^2 + c$, where a , b and c are integers.

Vertically dilated by a factor of 2:

$$\frac{y}{2} = x^2 - 8x + 17$$

$$= 2x^2 - 16x + 34$$

Reflected about y-axis:

$$y = 2(-x)^2 - 16(-x) + 34$$

$$= 2x^2 + 16x + 34$$

Completing the square

$$y = 2(x^2 + 8x + 17)$$

$$= 2((x+4)^2 + 1)$$

$$= 2(x+4)^2 + 2$$

10. Find the equation of the new function if $f(x) = x^3$ is transformed in the following order:

3

1. Horizontal dilation by a factor of 3
2. Reflection across the y-axis
3. Translated to the left by 1 unit

$$1, f\left(\frac{x}{3}\right) = \left(\frac{x}{3}\right)^3$$

$$2, f\left(-\frac{x}{3}\right) = \left(-\frac{x}{3}\right)^3 = -\frac{x^3}{27}$$

$$3, f\left(-\frac{(x+1)}{3}\right) = -\frac{(x+1)^3}{27}$$

11. Point $P(3, -5)$ lies on the function $y = f(x)$.

1

The function is transformed to $y = f(1-x) + 2$.

What are the coordinates of the image of P ?

① Horizontal Dilation - reflect over y-axis

$$P(x, y) = (3, -5)$$

$$P'(-x, y) = (-3, -5) \rightarrow 1 \text{ mark}$$

② Horizontal Translation - 1 unit right

$$P'(-(x-1), y) = (-2, -5) \rightarrow 1 \text{ mark}$$

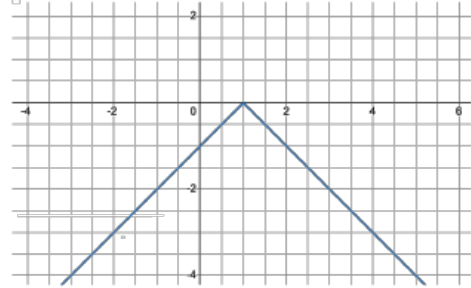
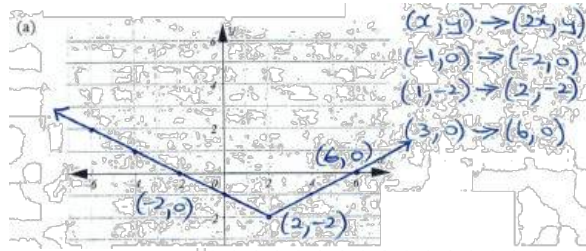
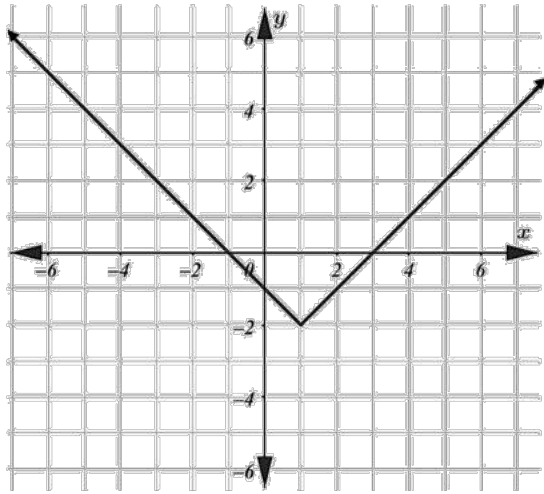
③ Vertical Translation - 2 units up

$$P'(-(x-1), y+2) = (-2, -3) \rightarrow 1 \text{ mark}$$

12. The graph of the function $y = f(x)$ has been shown below.

(a) Sketch the graph of $y = f\left(\frac{x}{2}\right)$ and show all intercepts.

(b) Sketch the graph of $y = -f(x) - 2$ and show all intercepts.



13. If $f(x) = x^2$ and $g(x) = -1 - (x + 3)^2$ what transformations map the graph of $f(x)$ to $g(x)$?

- A) Horizontally translated 3 units to the left, vertically translated 1 unit up and reflected across the x -axis
- B) Horizontally translated 3 units to the right, vertically translated 1 unit up and reflected across the x -axis
- C) Horizontally translated 3 units to the left, vertically translated 1 unit up and reflected across the y -axis
- D) Horizontally translated 3 units to the right, vertically translated 1 unit down and reflected across the x -axis

14. The curve $y = kx^2 + c$ is subject to the following transformations:

- Translated 2 units to the right.
- Dilated vertically by a factor of 4.
- Reflected in the y -axis.

The final equation of the curve is $y = 8x^2 + 32x - 8$. Find the values of k and c .

$$y = k(x - 2)^2 + c$$

$$\frac{y}{4} = k(x - 2)^2 + c$$

$$\frac{y}{4} = k(-x - 2)^2 + c$$

$$\frac{y}{4} = k(x + 2)^2 + c$$

$$y = 4k(x^2 + 4x + 4) + 4c$$

$$y = 4kx^2 + 16kx + 16k + 4c$$

$$4k = 8$$

$$\therefore k = 2$$

$$16k + 4c = -8$$

$$32 + 4c = -8$$

$$\therefore c = -10$$